

TASK 2: Access and print the element at a given index in an array

```
public class Main{
    public static void main(String[] args) {

        int[] numbers = {10, 20, 30, 40, 50};

        int index = 2;

        System.out.println(numbers[index]);
    }
}
```

TASK 3: Search for an element using binary search

```
public class BinarySearch{
    public static int binarySearch(int[] arr, int target) {
        int left = 0;
        int right = arr.length - 1;

        while(left <= right) {
            int mid = left + (right - left) / 2;

            if(arr[mid] == target) {
                return mid;
            } else if (arr[mid] < target) {
                left = mid + 1;
            } else {
                right = mid - 1;
            }
        }
        return -1;
    }

    public static void main(String[] args) {
        int[] arr = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91};
        int target = 23;
        int result = binarySearch(arr, target);
        System.out.println("Index:" + result);
    }
}
```

TASK 4: Find the maximum element in an array

```
public class MaxElement {
    public static int findMaximum(int[] arr) {
```

```

    if(arr == null || arr.length == 0) {
        throw new IllegalArgumentException("Array must not be empty");
    }

    int max = arr[0];

    for(int i = 1; i < arr.length; i++) {
        if(arr[i] > max) {
            max = arr[i];
        }
    }

    return max;
}

public static void main(String[] args) {
    int[] numbers = {12, 45, 98, 33, 27};
    System.out.println("Maximum element:" + findMaximum(numbers));
}
}

```

TASK 5:Find the Kth smallest element

```

import java.util.Arrays;
import java.util.Scanner;

public class SmallestElement {
    public static void Main(String[] args) {
        int[] numbers = {7, 10, 12, 15, 18};
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter K:");
        int k = sc.nextInt();
        Arrays.sort(numbers);

        if (k >= 1 && k <= numbers.length) {

            System.out.println("Sorted Array: " + Arrays.toString(numbers));

            System.out.println( k + "th smallest element = " + numbers[k - 1]);

        } else {
            System.out.println("Invalid K");
        }
        sc.close();
    }
}

```

TASK 6:Print all possible pairs of elements

```
public class PairsOfElements {  
    public static void main(String[] args) {  
        int[] numbers = {10, 20, 30, 40};  
        for (int i = 0; i < numbers.length; i++) {  
            for (int j = i + 1; j < numbers.length; j++) {  
                System.out.println( "(" + numbers[i] + ", " + numbers[j] + ")" );  
            }  
        }  
    }  
}
```